PARENT ONLY — SOLVED ANSWER KEY

Class 9 · Maths · Weekly Test 1

Sunday, 18 October 2026 · 60 minute · 40 marks · Test ID: T2026-10-19

Class 9 · Ch 1 Coordinates · Ch 2 Linear Polynomials · Ch 5 Circles (5.1–5.8)

Instructions: Saare prashn compulsory hain. Section E mein internal choice hai. · Har step likho. Figure jahan zaroori ho, banao. · Calculator allowed nahi hai.

Section A · MCQ + Assertion-Reason

Q1 [1] A point on the y-axis, 5 units below the origin, has coordinates: **1 mark**

a. (5, 0) b. (–5, 0)
c. (0, –5) d. (0, 5)

Expected answer / marking points: (c) (0, –5)

Q2 [1] The coordinates of every point on the x-axis are of the form: **1 mark**

a. (0, y) b. (x, 0)
c. (x, x) d. (x, y) with x = y

Expected answer / marking points: (b) (x, 0)

Q3 [1] Which of the following is a linear polynomial? **1 mark**

a. $x^2 + 1$ b. 5 – 4y
c. $x^3 - 2x$ d. 7

Expected answer / marking points: (b) 5 – 4y (degree 1)

Q4 [1] The lines y = 2x + 3 and y = 2x – 5 are: **1 mark**

a. perpendicular b. parallel
c. the same line d. intersecting at (0, 3)

Expected answer / marking points: (b) parallel — same slope 2, different y-intercepts.

Q5 [1] Linear decay is represented by a straight line with: **1 mark**

a. positive slope b. zero slope
c. negative slope d. no y-intercept

Expected answer / marking points: (c) negative slope

Q6 [1] An arc subtends an angle of 70° at the centre of a circle. The angle it subtends at a point on the remaining part of the circle is: **1 mark**

a. 140°
b. 70°
c. 35°
d. 110°

Expected answer / marking points: (c) 35° — half of the angle at the centre.

Q7 [1] ABCD is a cyclic quadrilateral with ∠A = 75°. Then ∠C = **1 mark**

a. 75°
b. 105°
c. 115°
d. 15°

Expected answer / marking points: (b) 105° — opposite angles of a cyclic quadrilateral add up to 180°.

Q8 [1] The angle subtended by a diameter at any point on the circle is: **1 mark**

a. 45°
b. 60°
c. 90°
d. 180°

Expected answer / marking points: (c) 90°

Q9 [1] How many circles can pass through three points that are not on one straight line? **1 mark**

a. None
b. Exactly one
c. Exactly two
d. Infinitely many

Expected answer / marking points: (b) Exactly one (the circumcircle).

Q10 [1] **A:** Equal chords of a circle are at equal distances from the centre. **1 mark**
R: Of two unequal chords of a circle, the longer chord is closer to the centre.

a. Both A and R are true, and R is the correct explanation of A.
b. Both A and R are true, but R is not the correct explanation of A.
c. A is true, but R is false.
d. A is false, but R is true.

Expected answer / marking points: (b) Both true, but R does not explain A.

Q11 [1] **A:** The point (4, –4) lies on the line y = –x. **1 mark**
R: For every point on the line y = –x, the x-coordinate and the y-coordinate are equal.

a. Both A and R are true, and R is the correct explanation of A.
b. Both A and R are true, but R is not the correct explanation of A.
c. A is true, but R is false.
d. A is false, but R is true.

Expected answer / marking points: (c) A true (–4 = –4); R false — on y = –x the coordinates are negatives of each other.

Section B · Very Short Answer

Q12 [2] You have ₹800 and save ₹250 every month. Write the amount after n months as a linear pattern and find the amount after (i) 6 months (ii) 2 years. **2 marks**

> **Expected answer / marking points:** A = 800 + 250n [1]. (i) n = 6: ₹2,300 [0.5]; (ii) n = 24: ₹6,800 [0.5].

Q13 [2] Find the distance of the point P(6, 8) from the origin. **2 marks**

> **Expected answer / marking points:** OP = $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ units (Pythagoras on the right triangle formed with the axes) [1+1].

Q14 [2] In a circle of radius 13 cm, a chord is 5 cm away from the centre. Find the length of the chord. **2 marks**

> **Expected answer / marking points:** Half-chord = $\sqrt{(13^2 - 5^2)} = \sqrt{144} = 12$ cm [1]; chord = 24 cm (perpendicular from centre bisects the chord) [1].

Q15 [2] PQRS is a cyclic quadrilateral with $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$. Find x, $\angle P$ and $\angle R$. **2 marks**

> **Expected answer / marking points:** $\angle P + \angle R = 180^\circ \Rightarrow 5x - 10 = 180 \Rightarrow x = 38$ [1]; $\angle P = 86^\circ$, $\angle R = 94^\circ$ [1].

Section C · Short Answer

Q16 [3] A positive number is 5 times another number. If 21 is added to both numbers, one of the new numbers becomes twice the other new number. Find the numbers. **3 marks**

> **Expected answer / marking points:** Let numbers be x and 5x [0.5]. 5x + 21 = 2(x + 21) [1] ⇒ 3x = 21 ⇒ x = 7 [1]. Numbers: 7 and 35 [0.5]. (Check: 56 = 2 × 28.)

Q17 [3] M(–7, 1) is the midpoint of the segment joining A(3, –4) and B(x, y). Find the coordinates of B. **3 marks**

> **Expected answer / marking points:** (3 + x)/2 = –7 ⇒ x = –17 [1.5]; (–4 + y)/2 = 1 ⇒ y = 6 [1.5]. B(–17, 6).

Q18 [3] Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle. The distance between them is 7 cm. Find the radius of the circle. **3 marks**

> **Expected answer / marking points:** Let d = distance of the 24 cm chord from centre; the 10 cm chord is at d + 7 [0.5]. $r^2 = 12^2 + d^2 = 5^2 + (d + 7)^2$ [1] ⇒ 144 = 74 + 14d ⇒ d = 5 [1] ⇒ $r^2 = 169$, r = 13 cm [0.5].

Q19 [3] For the line y = –3x + 4: (i) write its slope and y-intercept, (ii) find the point where it cuts the x-axis, (iii) is this linear growth or linear decay? **3 marks**

> **Expected answer / marking points:** (i) slope –3, y-intercept 4 [1]; (ii) 0 = –3x + 4 ⇒ (4/3, 0) [1]; (iii) linear decay — slope negative [1].

Section D · Case-Based

Q20 **4 marks** **[4]**

A computer screen is 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 pixels has its centre at A(100, 150). Another circular icon of radius 100 pixels has its centre at B(250, 230).

(i) Does any part of the circle with centre A go outside the screen? Justify. [1]
(ii) Does any part of the circle with centre B go outside the screen? Justify. [1]
(iii) Find the distance AB and decide whether the two circles intersect. [2]

Expected answer / marking points: (i) Leftmost x = 100 − 80 = 20, lowest y = 150 − 80 = 70; right 180, top 230 — all within 0–800 and 0–600, so fully inside [1].
(ii) Left 150, bottom 130, right 350, top 330 — fully inside [1].
(iii) AB = $\sqrt{(150^2 + 80^2)}$ = $\sqrt{28900}$ = 170 px [1]. Sum of radii = 180 > 170, so the circles intersect [1].

Section E · Long Answer

Q21 **5 marks** **[5]**

Prove that the angle subtended by an arc at the centre of a circle is twice the angle subtended by it at any point on the remaining part of the circle. Draw a neat figure.

OR

A growing pattern of hexagons is made with matchsticks; each new hexagon shares one side with the previous one (Stage 1 = 1 hexagon, Stage 2 = 2 joined hexagons, ...). (i) Make a table of matchsticks for stages 1 to 4. (ii) Find the rule for stage n. (iii) How many matchsticks are needed for stage 15? (iv) Can 200 matchsticks form a stage exactly? Justify.

Expected answer / marking points: Theorem: Figure with arc PQ, centre O, point R on the remaining circle; join RO and extend to S [1]. In △OPR, OP = OR ⇒ ∠OPR = ∠ORP; exterior ∠POS = ∠OPR + ∠ORP = 2∠ORP [1.5]. Similarly ∠QOS = 2∠ORQ [1]. Adding: ∠POQ = 2(∠ORP + ∠ORQ) = 2∠PRQ [1.5].
OR (matchsticks): Table 6, 11, 16, 21 [1.5]; rule 5n + 1 [1.5]; stage 15: 76 [1]; 5n + 1 = 200 ⇒ n = 39.8, not a whole number, so no [1].

Parent copy — keep separate from the student paper.